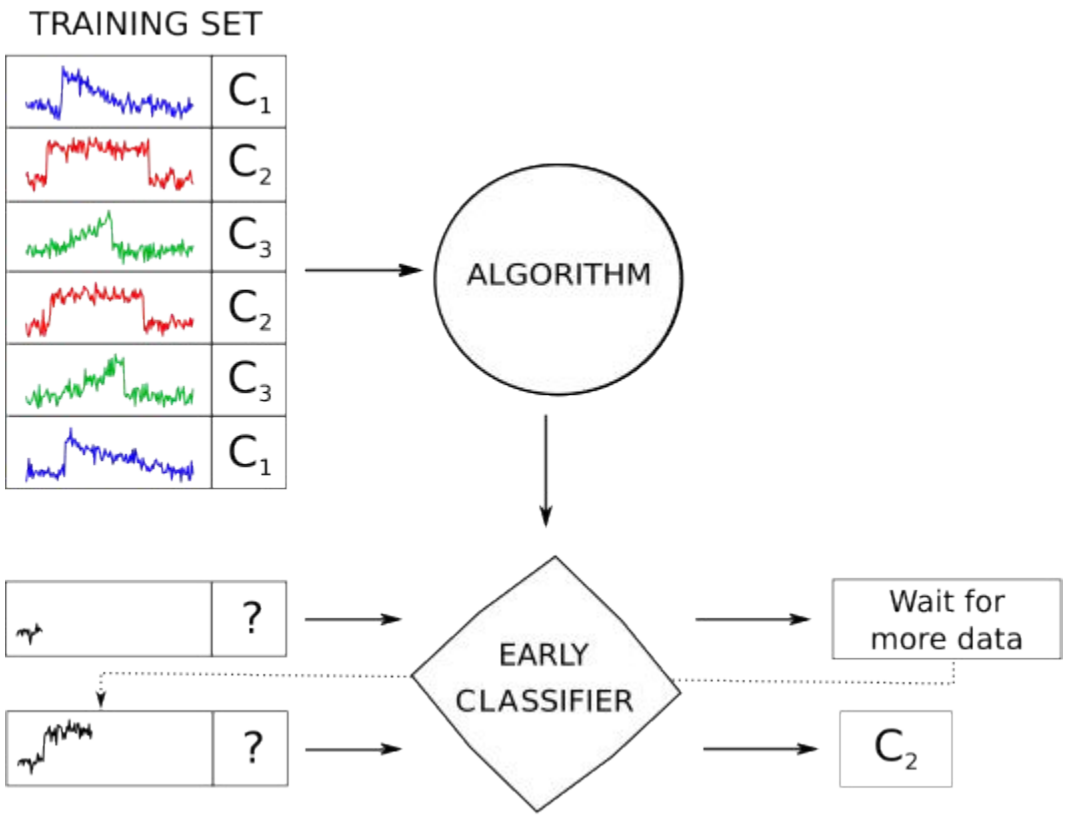
EARLY CLASSIFICATION OF TIME SERIES


TRAINING SET
C1
C2
C3
C2
C3
C1
ALGORITHM
?
?
EARLY CLASSIFIER
Wait for more data
C2

Erlanger Health System
Clinical Engineering

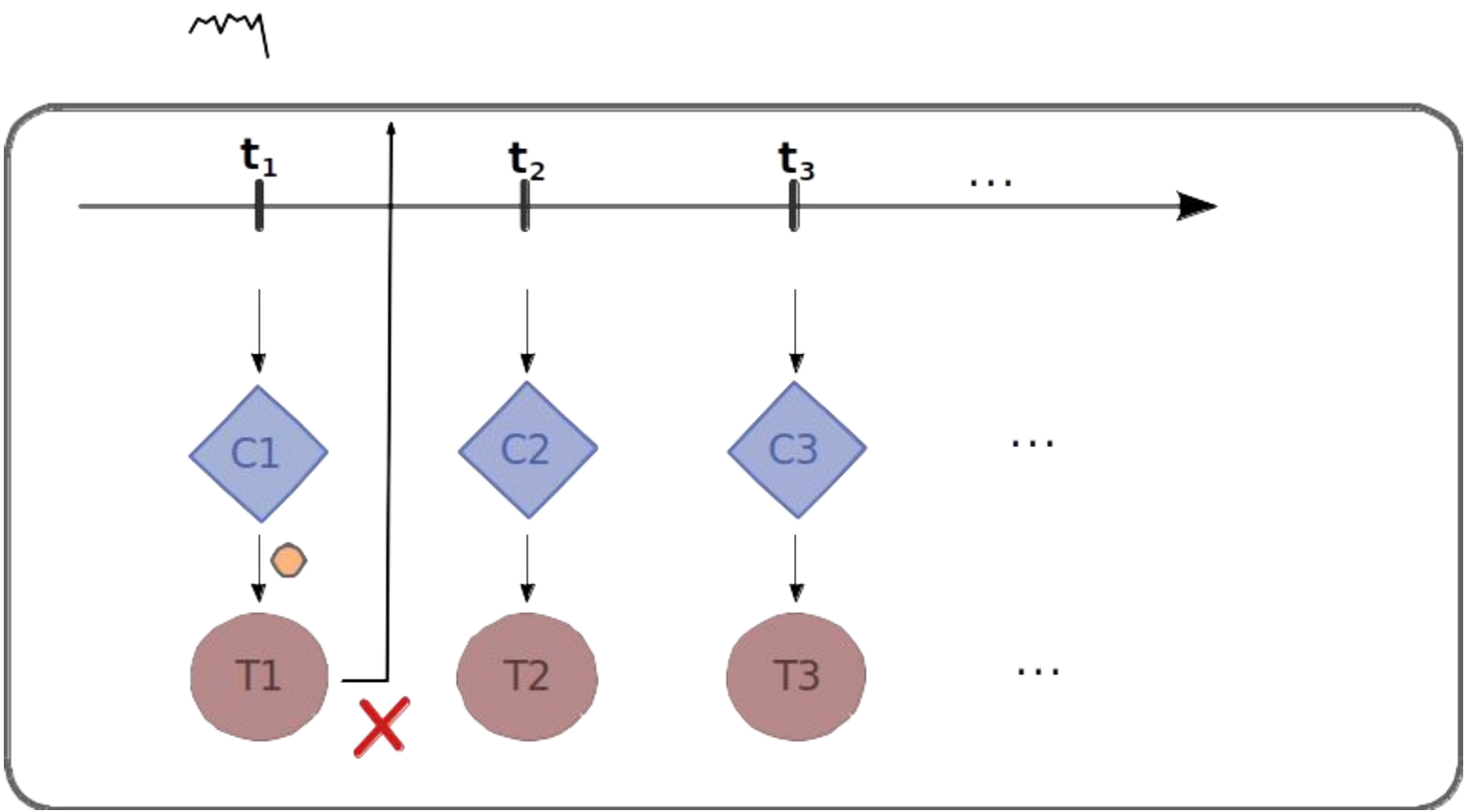
Windows Security
Windows Security
Threats found
Microsoft Defender Antivirus found threats. Get details.
Dismiss
ENG
UK
11:37
21/01/2022

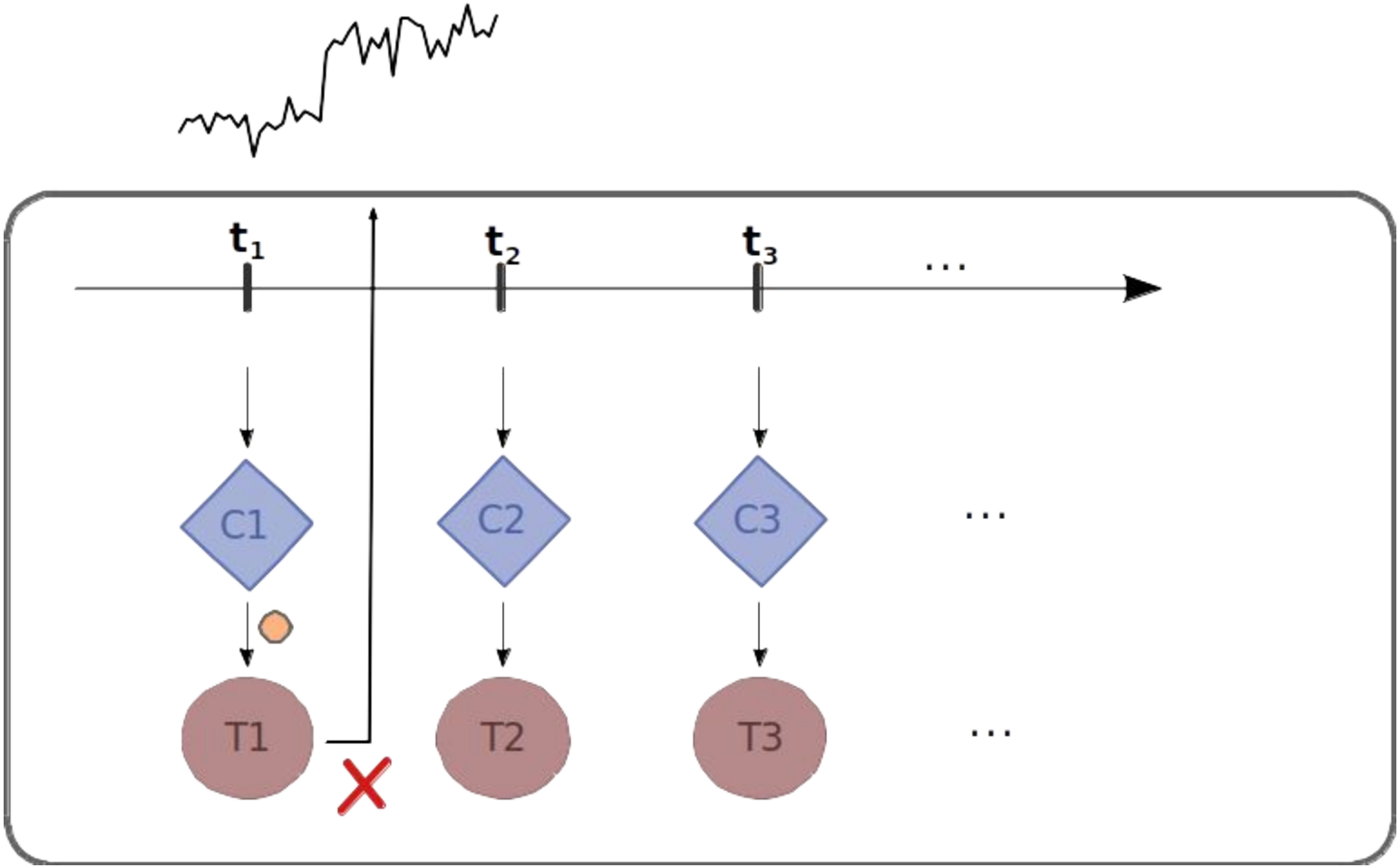
KUKA

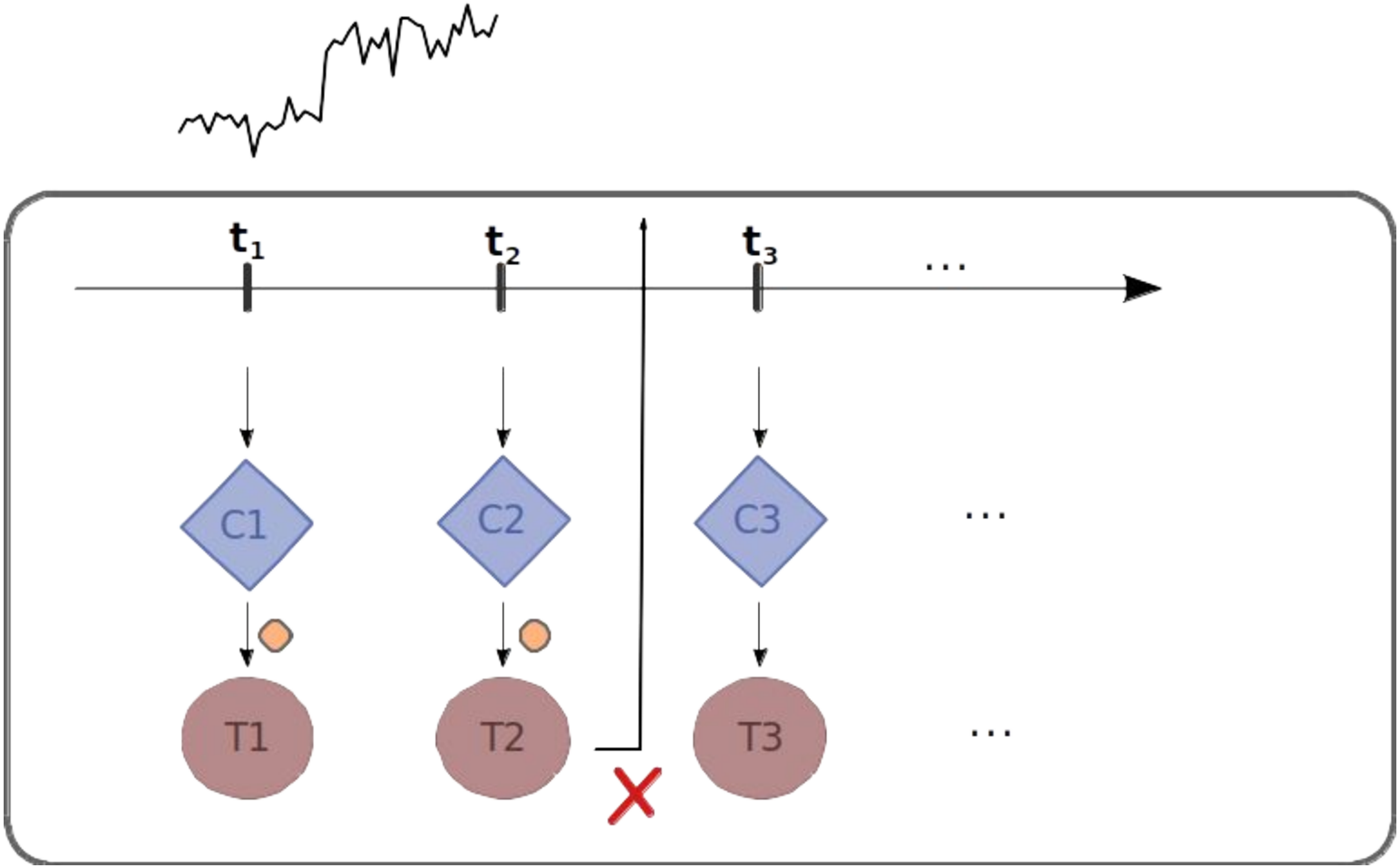
ACCURACY VS. EARLINESS

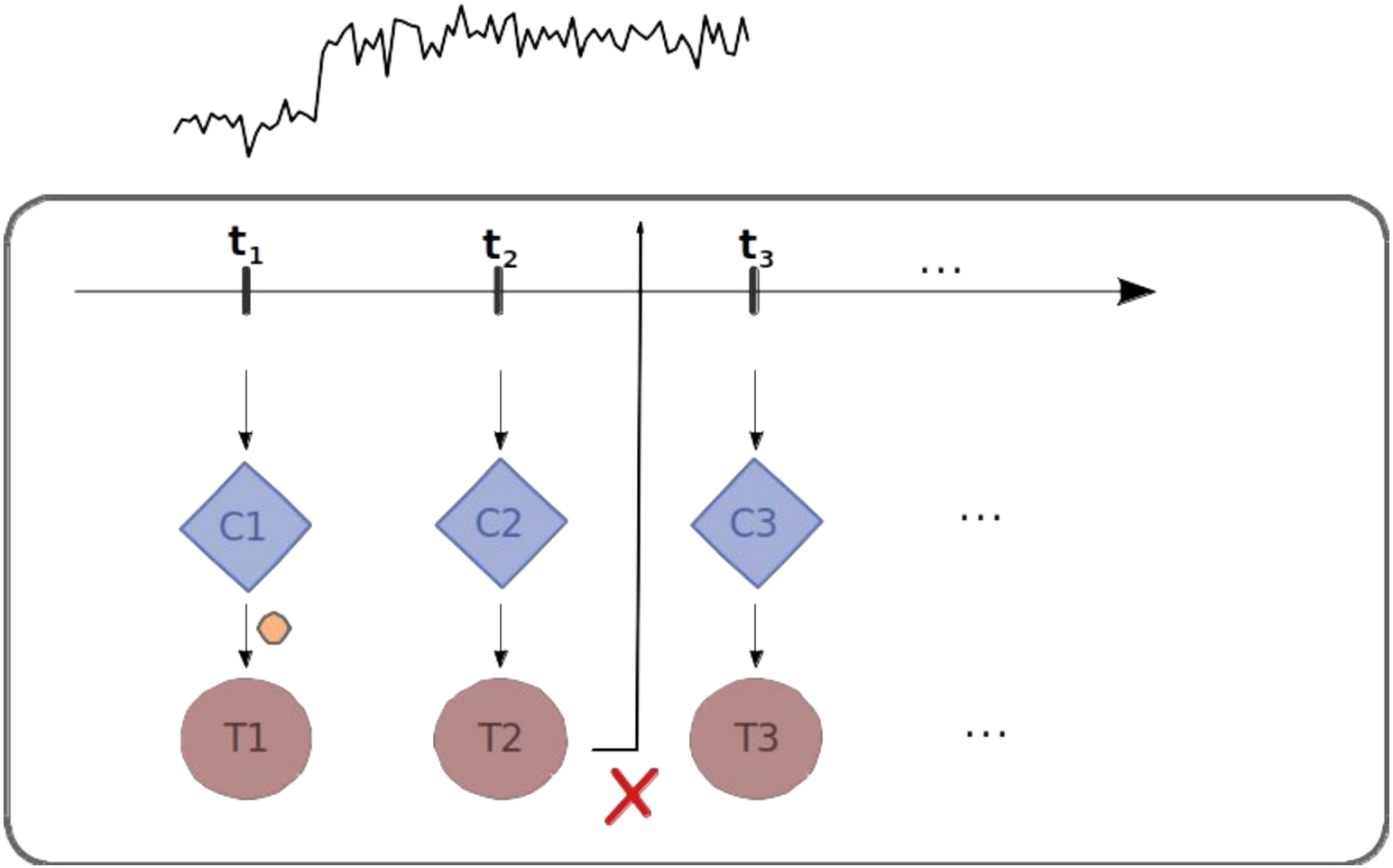
STATE-OF-THE-ART

- Based on studying the stability of the 1NN relationships over time.
- Combine different classifiers and design reliability conditions based on obtained predictions.
- Based on finding shapelets that appear “early” in time.
- Based on optimized trigger functions.

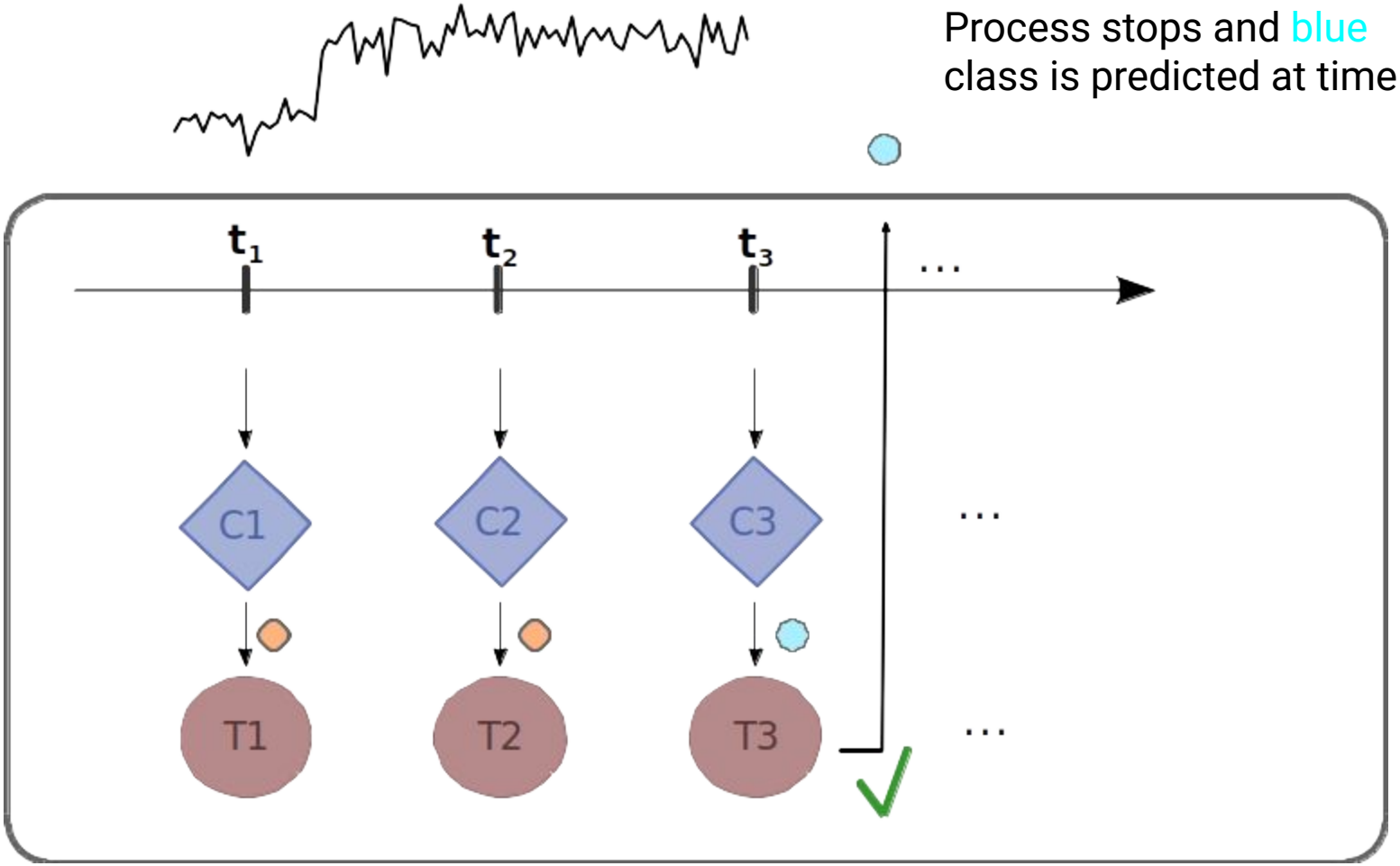

t1
t2
t3
...
C1
C2
C3
...
T1
T2
T3
...


t1
t2
t3
...
C1
C2
C3
...
T1
T2
T3
...


t1
t2
t3
...
C1
C2
C3
...
T1
T2
T3
...


t1
t2
t3
...
C1
C2
C3
...
T1
T2
T3
...

Process stops and blue class is predicted at time t_3



Example Trigger Function

$$s(p_1^t, p_2^t, \ldots, p_k^t, t) = \begin{cases} 0 & if \quad \gamma_1 p_{1:k}^t + \gamma_2 (p_{1:k}^t - p_{2:k}^t) + \gamma_3 \frac{t}{L} + \gamma_4 \leq 0 \\ 1 & \text{otherwise} \end{cases}$$

- Input:
  - Posterior class probabilities obtained by the classifier at time t
  - Timestamp t
- Output:
  - 0 (stop and make prediction)
  - 1 (wait for more data)

Cost Functions and Optimization Problem

- $C_a(x, \gamma)$ missclassification cost, typically the 0-1 loss.
- $C_e(x, \gamma)$ earliness cost, a function that increases with time, typically a linear function.

$$min_\gamma(C_a(X, \gamma), C_e(X, \gamma))$$

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